FACULTY OF COMPUTING

SEMESTER II

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SECP 3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02

Lab Tutorial 1: Microsoft Azure

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1.0 Project Overview

Many companies struggle with "data silos," where important info like sales records is stuck in old on-premise systems. This makes it hard to analyze data, such as how gender affects buying habits. Our goal is to fix this by moving data to the cloud. Hence, this project creates a "Single Source of Truth." It gives stakeholders a simple interface to query data, helping them make better decisions on marketing and inventory.

Objectives:

- **Cloud Migration:** Securely migrate legacy data from an on-premises SQL Server to the Microsoft Azure cloud environment.
- **Automation:** Build a robust, scalable, and automated data pipeline for ingestion, transformation, and storage.
- **Business Intelligence:** Develop a high-impact Power BI dashboard to visualize KPIs, including sales by gender, product category, and total revenue.

2.0 Description of the solution

The solution implements a modern Medallion Architecture (Bronze, Silver, and Gold layers) to ensure data integrity and quality as it moves through the pipeline.

1. **Ingestion**

   The data is extracted into Azure Data Factory (ADF) with a Self-Hosted Integration Runtime to securely extract data from the on-premise SQL Server.

2. **Storage (Data Lake)**

   The extracted data is landed in Azure Data Lake Storage (ADLS) Gen2.

   - **Bronze Layer:** Data is stored in its native, raw format to preserve the original state of the information.

3. **Transformation**

   The data is proceed with utilising Azure Databricks (or Synapse Analytics) to process the data through the remaining stages:

   - **Silver Layer:** Data is cleaned, filtered, and standardized.
   - **Gold Layer:** Data is aggregated and enriched according to specific business logic, making it query-friendly for final analysis

4. **Loading & Serving**
   The refined "Gold" data is loaded into Azure Synapse Analytics or an Azure SQL Database. This serves as the high-performance serving layer for the final analytics engine.
5. **Visualisation**
   Power BI connects to the served data, allowing users to view key metrics and get insights.

3.0 Technology Used

In order to fulfill the project requirements, the following Azure services below are utilised, where This stack will ensure the pipeline is functional and follows the enterprise best practices.

Table 1: Azure Services Used

Technology	Role	Description
Azure Data Factory	Orchestration	Act as the central hub to orchestrate the data movement and create the automated pipeline.
Azure Data Lake Gen2	Storage Layer	Serves as the scalable storage layer for raw and processed data.
Azure Databricks	Transformation Engine	A Spark-based environment utilized for high-performance data cleaning and complex transformations of raw data.
Azure SQL Database	Relational Storage	Serves as both the simulated on-premise source and the final destination for processed, query-ready data.
Azure Key Vault	Security Manager	Manages secrets, credentials, and connection strings to ensure the pipeline follows strict security protocols.

Microsoft Power BI	Visualisation	Connecting MS Power BI to the curated layer in Azure to build the dashboard for stakeholder analysis.

4.0 Data Pipelines Architecture

The implementation of the project is based on end-to-end cloud data platform architecture as shown in Figure 1 below:

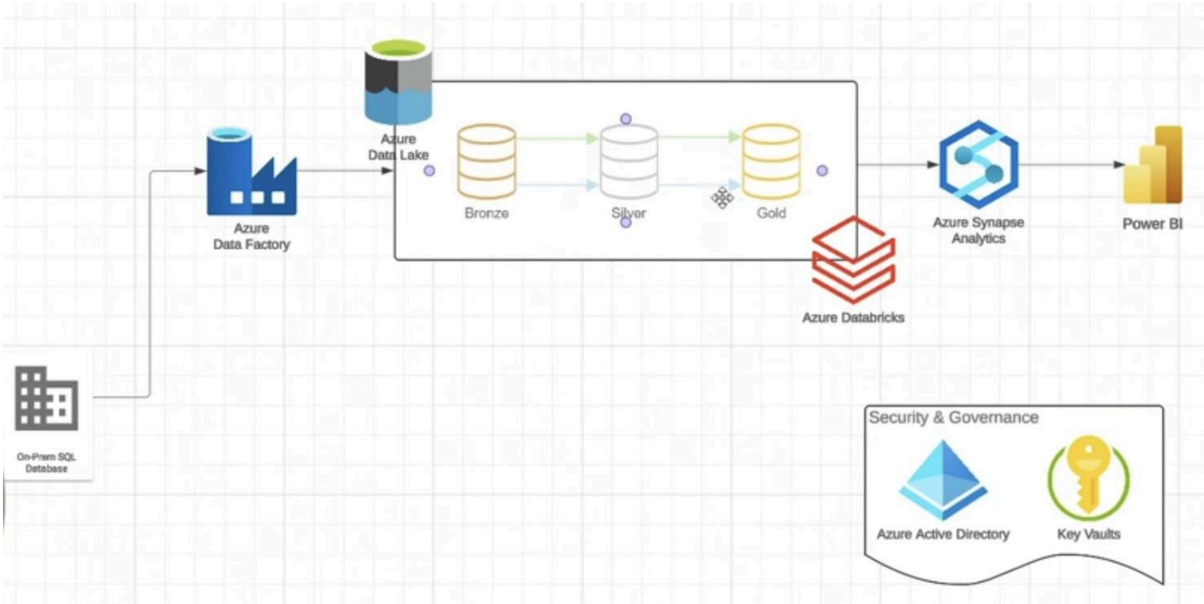

Figure 1: Data Pipeline Architecture

The project uses a typical End-to-End Cloud Data Platform architecture. This framework ensures a systematic flow of data from raw to extraction to business intelligence.

- Data Source: The old data is stored in a Microsoft SQL Server on-premise.
- Orchestration & Ingestion: Azure Data Factory (ADF) serves as the brains, transferring information into the cloud with a Self-Hosted Integration Runtime of secure connectivity.
- The Medallion Strategy: Data is stored in the Azure Data Lake storage (ADLS) Gen2 in three functional areas:
    - Bronze: raw and immutable landing zone,
    - Silver: standardized and filtered data using Azure Databricks.
    - Gold: Premium, aggregated tabular data prepared to be used in business.
- Serving & Consumption: Data is processed in Azure Synapse Analytics (Serverless SQL) and displayed using Microsoft Power BI.

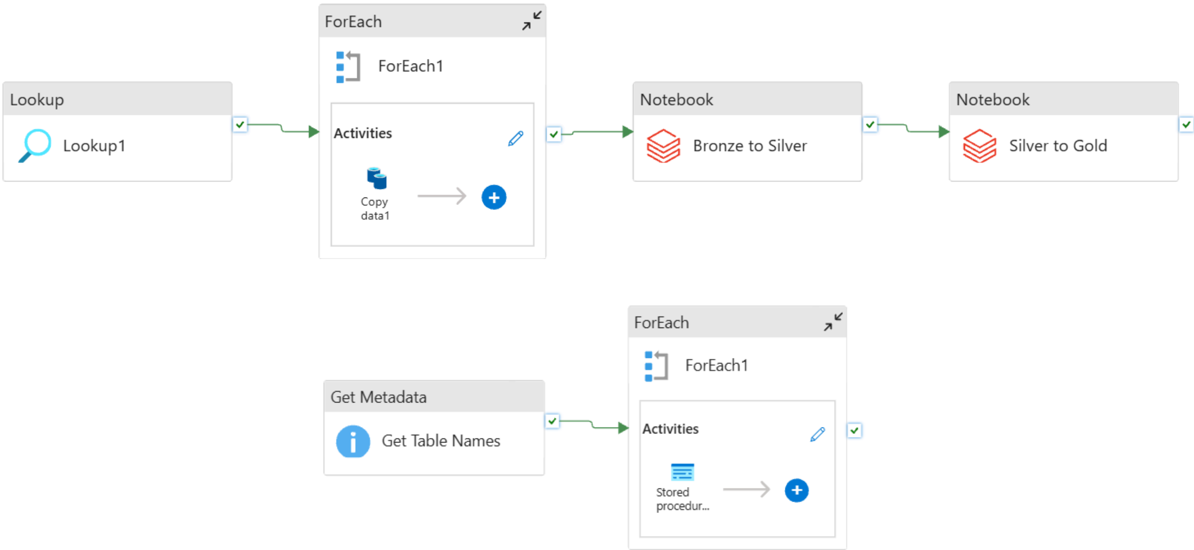

Figure 2: Pipeline Architecture in Azure

The architecture in Figure 2 will describe the exact logic and operations that we used Azure Data Factory to implement the above architecture.

- **Metadata-Driven approach**
  - Metadata Discovery: It will use a 'Get Metadata' activity to scan the source SQL database and dynamically obtain a list of existing table names at runtime.
  - Iterative Processing: A 'ForEach' container is set to repeat over the metadata list. This reasoning can allow the pipeline to automatically process new tables introduced into the source system without having to reconfigure the pipeline structure.
- **Layer-to-Layer Orchestration**
  - Integrity Checks: A 'Lookup' is performed at the beginning of the sequence of transformations. This checks the parameters of sources and makes sure that the environment has the required requirements prior to the start of processing.
  - Notebook Execution: Azure Databricks Notebook activities are created as a part of the ADF pipeline. These actions invoke PySpark codes that handle the elaborate logic needed to transfer the data between Bronze and Silver layers to Gold layers.
  - Security & Governance: The pipeline will be connected to the Azure Key Vault to improve security posture. Instead of storing sensitive credentials in the ADF connection strings, the pipeline reads secrets during execution. This deployment will make sure that the passwords of databases and services keys

are never revealed in any logs of the pipeline or source code, thus meeting the requirements of enterprise-level security.

5.0 Microsoft Power BI Dashboard

The dashboard developed as shown in Figure 3 and Figure 4 as a part of this tutorial gives a detailed overview of the sales lifecycle.

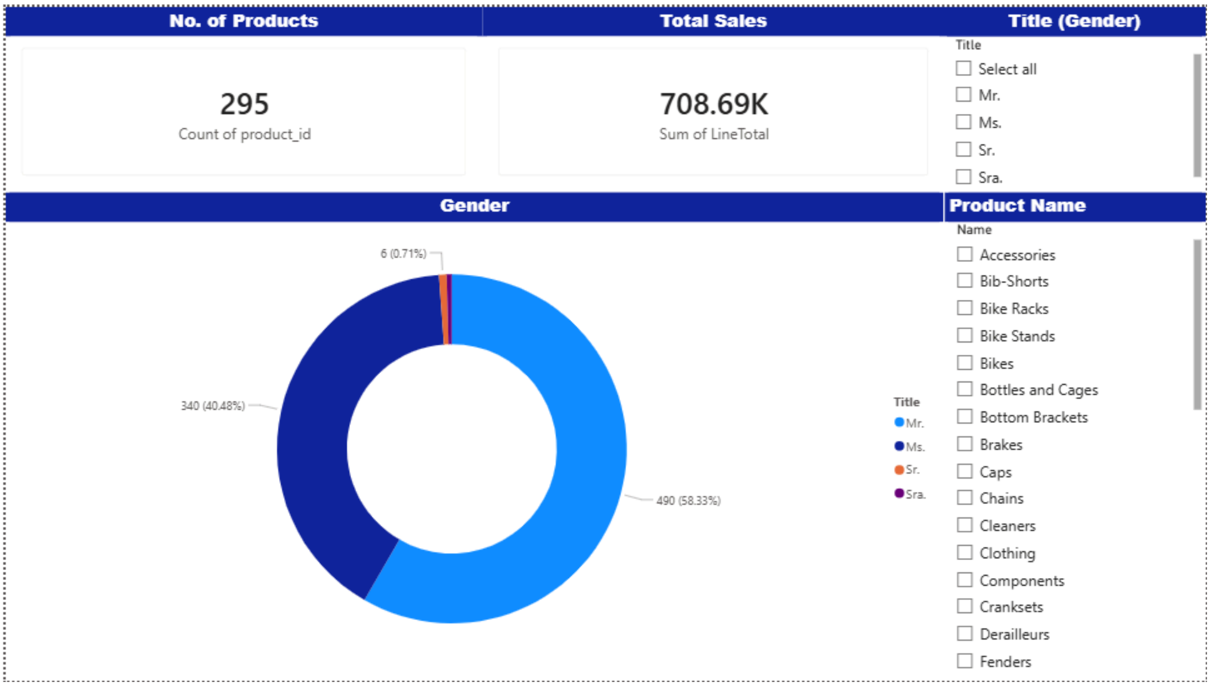

Figure 3: Strategic Sales Insights & Market Share Dashboard

The Dashboard is a strategic summary tool, which gives high level visibility on the volume of sales and the type of customer that the organization has. It also has a high-quality Key Performance Indicator (KPI) section, which emphasizes the Total Number of Products (295) and Total Sales revenue (708.69K), which provides the Total Number of the size of the business and financial activity. The analysis will focus on a Gender Distribution donut chart whereby the customer base in terms of titles will be segmented to show that the largest portion of the market share will be represented by categories of Mr. and Ms. with the percentage of 58.33 and 49.48 respectively.

The functionality of the dashboard is further supported by dual Slicers one is Title (Gender) Slicer and the other is Product Categories Slicer that allows the core metrics to dynamically be filtered. This interactive construction enables users to dig into individual product lines to determine the contribution to the total revenue and demographic breakup by individual category. It allows the specific marketing and inventory optimization measures.

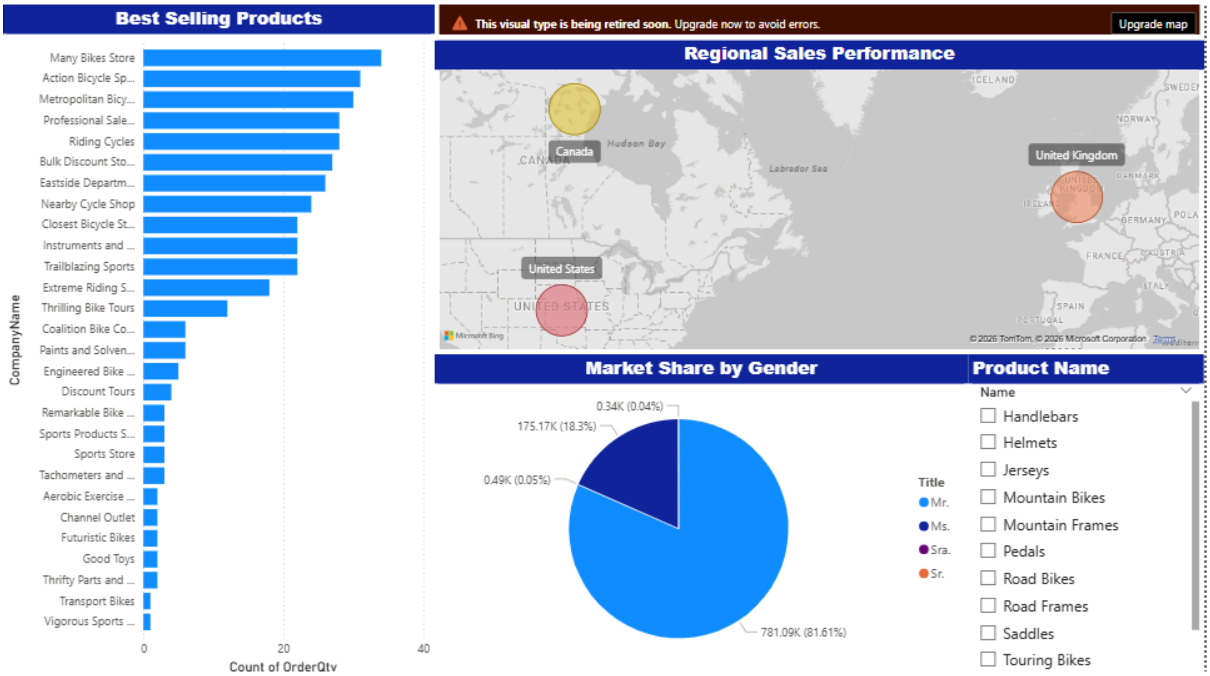

Figure 4: Global Revenue Analytics Dashboard

The Global Revenue Analytics Dashboard is a centralized analytical dashboard that converts data on the Gold tier into actionable business intelligence using three main dimensions: Geospatial Performance, Product Popularity and Customer Demographics. With a Relational Data Model, the dashboard also supports Dynamic Cross-Filtering, which enables users to click on certain areas on the Product Slicer and immediately refresh the Best Selling Products bar chart and the Market Share pie chart. This interconnected design enables stakeholders to perform real-time, multi-dimensional analysis, moving from high-level global sales overviews to specific demographic trends with a single click.

6.0 Reflection

1. **Nur Firzana Binti Badrus Hisham**

   This tutorial has opened my eyes to the real world of data engineering. A huge leap forward is leaving files on a manual transfer to letting the cloud do everything automatically. The experience of creating an Azure Data Factory is very challenging, to fetch the data itself, and then cleaning it up in Databricks made me realize why I cannot just use the raw data but must first clean it first to be useful. I also learned that security is not an additional procedure and the idea to use Azure Key Vault to hide passwords made the entire project look much more professional and secure. It was most enjoyable to see it all in Power BI. It is pretty cool how a bunch of messy tables from a local server can turn into a clean, interactive dashboard that actually helps people make business decisions. Overall, I feel like I finally get how to build a solid, secure system that does not just store data but actually makes sense of it.

2. **Nuraisyah Binti Mohd Zikre**

   Throughout the whole process of completing this tutorial, I’ve encountered many kinds of errors in order to successfully complete the tutorial. This experience allows me to learn and gain many kinds of knowledge about data pipelines. Moreover, this tutorial allows me to explore a lot of Microsoft Azure stack services in building pipelines for data visualisation. Besides, I’ve also learnt how each different Azure services used different syntax, and how the syntax connects all tables into one schema after importing it in Microsoft Power BI for the data visualisation. Not to mention that this experience allows me to use Microsoft Power BI for the development of the data visualisation. The experience of using Microsoft Power BI almost the same as using Tableau.

3. **Wu Chen Xi**

   Building this end-to-end cloud data platform was a challenging journey that taught me much more than just technical skills. I started from scratch and faced several significant hurdles. Initially, I struggled with my local environment, dealing with SQL Server installation errors and Java configuration issues that prevented the Self-Hosted Integration Runtime from working. I also hit a wall with cloud security; I realized that simply being an "Owner" was not enough to access secrets, so I had to learn how to configure Role-Based Access Control and Managed Identities properly. I even faced data corruption issues by mixing file formats and lost pipeline logic because I forgot to publish my work. These failures forced me to become a better troubleshooter.

However, overcoming these obstacles gave me deep practical knowledge. I successfully built a Medallion Architecture, moving data from Bronze to Gold layers. I moved beyond manual work by using Azure Data Factory's "Lookup" and "ForEach" loops with dynamic expressions like @item() to automate the ingestion of multiple tables. I learned to secure my pipelines by using Azure Key Vault instead of hard-coding passwords. In Databricks, I used PySpark to clean data and rename columns efficiently. Finally, I utilized Synapse Serverless SQL to query data directly from the lake. This project proved to me that while coding is important, the ability to debug errors and manage details is what truly makes a data engineer successful.